

∴ Maths:-

∴ MCQs:-

1. B

2. C

3. A

4. D

5. A

6. B

7. A

8. D

9. C

10. B

11. C

12. D

13. B

14. A

15. C

16. ~~A~~ D

17. C

18. C

19. B

20. P

21. C

22. A

23. D

24. A

$81m^4$

$(9m^2)$

$9m^2$

$(3m)^{-}$

$2x$

3

$2x$

$2x$

$2x$

$2x$

3

3

3

3

3

3

3

3

3

3

3

3

3

3

3

3

3

Q: 2(i)

(a)

$$\frac{x^3 \times x^{-2}}{x^4}$$

$$\frac{x^{3+(-2)}}{x^4}$$

$$\frac{x^1}{x^4}$$

$$x^{3-4}$$

$$\frac{1}{x}$$

(b) $(81)^{\frac{3}{4}}$

$$(3^4)^{\frac{3}{4}}$$

$$(3)^3 = 27$$

(c) $\sqrt[4]{a^7}$

$$(a^7)^{\frac{1}{4}}$$

$$a^{\frac{7}{4}}$$

Q: 2(ii)

$$\log 48$$

$$\log 6 \times 8$$

$$\log 3 \times 2 \times 2 \times 2 \times 2$$

$$\log 3 + \log 2 + \log 2 + \log 2 + \log 2$$

$$0.4771 + 0.3010 + 0.3010 + 0.3010 + 0.3010$$

$$= 1.6811$$

Q: 2(iii) $|2z-3| = 9$

$$2z-3=9 \quad \text{or} \quad 2z-3=-9$$

$$2z=9+3 \quad \text{or} \quad 2z=-9+3$$

$$2z=12 \quad \text{or} \quad 2z=-6$$

$$2z = \frac{6}{2} \quad ; \quad z = \frac{-6}{2}$$

$$z = 3, \quad z = -3$$

Check :-

$$2z - 3 = +9, \quad 2z - 3 = -9$$

$$2(6) - 3 = 9, \quad 2(-3) - 3 = -9$$

$$12 - 3 = 9, \quad -6 - 3 = -9$$

$$9 = 9, \quad -9 = -9$$

Q.3(i)

$$(a) \quad \frac{1+4n^2}{(1)^2 + (2n)^2 + 2(1)(2n) - 2(1)(2n) - (2n)^2}$$

$$(1+2n^2) - \cancel{4n^2} - (2n)^2$$

$$\frac{(1+2n^2-2n^2)(1+2n^2+2n)}{(2n^2-2n+1)(2n^2+2n+1)}$$

$$(b) \quad 49p^2 - 84pq + 36q^2$$

$$\therefore (a-b)^2 = (a)^2 - 2(a)(b) + (b)^2$$

$$a^2 - 2ab + b^2$$

$$(7p)^2 - 2(7p)(6q) + (6q)^2$$

$$(7p-6q)^2$$

$$(c) \quad \frac{x^2-64}{x^2+11x+24} \div \frac{x^2-16x+64}{x^2+3x}$$

$$\frac{(x)^2 - (8)^2}{x^2+11x+24} \div \frac{x^2-16x+64}{x(x+3)}$$

$$\frac{(x-8)(x+8)}{x^2+11x+24} \times \frac{x(x+3)}{x^2-16x+64}$$

$$\frac{(x-8)(x+8)}{x^2+3x+8x+24} \times \frac{x(x+3)}{\cancel{x^2-8x-8x+64}}$$

$$= \frac{(x-8)(x+8)}{(x^2-2(x)(8)+(8)^2)}$$

$$\frac{(x-8)(x+8)}{x(x+3)+8(x+3)} \times \frac{x(x+3)}{x(x-8)+8(x-8)}$$

$$\frac{(x-8)(x+8)}{x(x+3)+8(x+3)} \times \frac{x(x+3)}{(x-8)^2}$$

$$\frac{(x-8)(x+8)}{\cancel{x(x+3)}(x+8)\cancel{(x+8)}} \times \frac{x\cancel{(x+3)}}{(x-8)^2}$$

$$= \frac{x}{x-8} \quad \underline{\text{Ans}}$$

Q.3(ii)

$\{A(-1, -2), B(-1, 4), C(7, 4)\}$

(a) d_{AB}

$$= \sqrt{(-1 - (-1))^2 + (4 - (-2))^2}$$

$$= \sqrt{(-1 + 1)^2 + (4 + 2)^2}$$

$$= \sqrt{(0)^2 + (6)^2}$$

$$= \sqrt{0 + 36}$$

$$= \sqrt{36} \Rightarrow 6$$

$$(ii) d_{BC} :- \sqrt{(7 - (-1))^2 + (4 - 4)^2}$$

$$= \sqrt{(7 + 1)^2 + (0)^2}$$

$$= \sqrt{(8)^2 + 0}$$

$$= \sqrt{64}$$

$$= 8$$

$$\begin{aligned}
 d_{AC} &= \sqrt{(7-1)^2 + (4-(-2))^2} \\
 &= \sqrt{(7+1)^2 + (4+2)^2} \\
 &= \sqrt{(8)^2 + (6)^2} \\
 &= \sqrt{64 + 36} \\
 &= \sqrt{100} \\
 &= 10
 \end{aligned}$$

Pythagoras theorem

$$\begin{aligned}
 AB^2 + BC^2 &= AC^2 \\
 6^2 + 8^2 &= 10^2 \\
 36 + 64 &= 100 \\
 100 &= 100
 \end{aligned}$$

Therefore, $\triangle ABC$ is a right triangle, and the longest side AC is the hypotenuse.

② midpoint of AC {hypotenuse}

$$\begin{aligned}
 m &= \left\{ \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right\} \\
 &= \frac{-1 + 7}{2}, \frac{-2 + 4}{2} \\
 &= \frac{6}{2}, \frac{2}{2} \\
 &= 3, 1
 \end{aligned}$$

Now we'll show you that M.
is equally distant from all
these vertices...

$$M_A = \sqrt{(3 - (-1))^2 + (1 - (-2))^2}$$

$$= \sqrt{(3+1)^2 + (1+2)^2}$$

$$= \sqrt{(4)^2 + (3)^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25} \Rightarrow 5 \text{ units}$$

$$M_B = \sqrt{(3 - (-1))^2 + (1 - 4)^2}$$

$$= \sqrt{(3+1)^2 + (-3)^2}$$

$$= \sqrt{(4)^2 + (-9)}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25} \Rightarrow 5 \text{ units}$$

$$M_C = \sqrt{(3 - 7)^2 + (1 - 4)^2}$$

$$= \sqrt{(-4)^2 + (-3)^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25} \Rightarrow 5 \text{ units}$$